SIFAT ABDUL BARI

Mechanical Engineering | Computational Materials Science | High Entropy Alloys

Email Website Google Scholar

Research Interests

Advanced Materials Design | High-Entropy Alloys (HEAs) | Computational Materials Science | Mechanics of Materials | Atomistic Modelling | Molecular Dynamics (MD) | Density Functional Theory (DFT) | Machine-Learned Interatomic Potentials

Education

Master of Science in Mechanical Engineering — CGPA: 3.92/4.00
Islamic University of Technology (IUT), Board Bazar, Gazipur, Bangladesh
2023–Present — Courses Completed – Thesis Ongoing

Bachelor of Science in Mechanical Engineering — CGPA: 3.87/4.00
Islamic University of Technology (IUT), Board Bazar, Gazipur, Bangladesh
2019–2023 — 2nd out of 82 Students

Technical Skills

Molecular Dynamics (MD) Simulation	LAMMPS, OVITO, ATOMSK, VESTA, Quantum ESPRESSO
Programming and Computing	Python, C++, MATLAB, Arduino IDE
Machine Learning and Data-driven Modelling	Artificial Neural Network (ANN), Random Forest (RF), XGBoost
3D Modelling and Engineering Simulation	SOLIDWORKS, ANSYS Workbench, ANSYS Fluent
Post-processing and Data Visualization	Origin, Python, MATLAB
Scientific Writing and Documentation	LaTeX, Microsoft Office Suite
Image Processing and Illustration	Adobe Illustrator, Inkscape
Experimental Equipment and Testing	Universal Testing Machine (UTM), FDM 3D Printer, Oscilloscope, Beam Apparatus, Torsion Testing, Drop Testing, Impact Testing, Fatigue Testing
Languages	Bangla (Native), English (IELTS Overall 7.5)

Publications

Journal Articles

1. S. A. Bari, M. Fuad, K. F. Labib, M. Monjurul Ehsan, Y. Khan, and M. M. Hasan, "Enhancement of thermal power plant performance through solar-assisted feed water heaters: An innovative repowering approach," *Energy Conversion and Management: X*, vol. 22, p. 100550, 2024. doi: 10.1016/j.ecmx.2024.100550.

2. S. A. Bari, M. Fuad, M. Shahed Mahamud, A. A. Bhuiyan, and M. Ahiduzzaman, "Drying kinetics and energy-exergy performance of a hybrid solar-biomass dryer," Energy Convers. Manag. X, vol. 30, p. 101644, 2026, doi: 10.1016/j.ecmx.2026.101644.

Conference Articles

1. S. Abdul Bari, C. S. Alam, A. S. Shuvo, S. A. R. Khan, and M. S. Rahman, "A Molecular Dynamics Study of Shear Driven Solidification and High Temperature Mechanical Properties of Al0.3CoCrFeNi High Entropy Alloy," 2025, vol. Volume 2:, p. V002T03A089. doi: 10.1115/IMECE2025-166710.

Research Experience

Master's Thesis

Molecular Dynamics Study of Shear-Assisted Microstructural Evolution and Property Response in $Al_{0.3}CoCrFeNi$ High-Entropy Alloy

- Investigated the influence of shear-assisted solidification conditions on the microstructural evolution of $Al_{0.3}CoCrFeNi$ HEA using molecular dynamics simulations.
- Characterized processing-induced microstructural features, including grain refinement, twin boundary evolution, fivefold twin formation, and chemical short-range ordering.
- Correlated shear-tailored microstructures with tensile strength, high-temperature mechanical response, and deformation mechanisms.
- Evaluated cyclic loading behavior by analysing dislocation evolution and structural stability under repeated mechanical loading.
- Quantified chemical short-range order (CSRO) using Warren–Cowley parameters to understand local chemical ordering effects.
- Performed nanoindentation simulations to investigate hardness and subsurface defect evolution.
- Assessed primary irradiation damage tolerance of shear-engineered microstructures through molecular dynamics collision cascade simulations.
- Utilized LAMMPS and OVITO for atomistic modelling, defect analysis, and structure–property correlation.

Undergraduate Thesis

Design and Evolution of a Novel Solar Biomass Hybrid Dryer

- Designed and fabricated an advanced hybrid solar dryer after extensive literature review, design analysis, and experimental evaluation.
- Obtained drying parameters through experimental investigation under different operating modes.
- Evaluated thermodynamic performance through energy and exergy analysis.
- Determined energy and exergy efficiency for three different drying configurations.

LAMMPS-based Molecular Dynamics Projects

1. WMoZrTiTa Refractory High Entropy Alloy (RHEA)

- Performed MC/MD Hybrid simulations for microstructural evolution and mechanical property analysis of WMoZrTiTa RHEA.
- Investigated the influence of chemical short-range order (CSRO) on mechanical properties and deformation mechanisms.
- Performed radiation damage simulations using Primary Knock-on Atom (PKA) collision cascades.

2. Transition Metal Dichalcogenides (TMDs)

- Constructed monolayer models of h-BN, MoS_2, and WSe_2 using VESTA and ATOMSK.
- Evaluated tensile properties and analysed crack propagation behaviour.
- Investigated deformation mechanisms and nanoindentation-induced hardness response.

Academic and Professional Experience

Lecturer

Department of Mechanical and Production Engineering
Islamic University of Technology, Bangladesh
August 2023–Present

- Conducted theoretical courses and laboratory sessions in Machine Design, Mechanics of Materials, Statics and Dynamics, Thermodynamics, Fluid Mechanics, and Manufacturing-related courses.
- Prepared course files and documentation for Outcome Based Education (OBE) accreditation.
- Managed examination scheduling, class routines, result preparation, and academic documentation.
- Developed laboratory manuals and supervised laboratory activities.
- Advised undergraduate students and supported academic project development.

Student Supervision

- Co-supervised undergraduate thesis on "Mechanical Properties and Cracking Mechanism in Pristine and Defected WSe_2 2D TMD: A MD Study".
- Co-supervised undergraduate thesis on "MD Study of Deformation Mechanism and Mechanical Properties of AlHfNbTaTiZr Refractory High Entropy Alloy".
- Supervised measurement, instrumentation, and control projects involving sensor-based systems, data acquisition, and feedback-controlled mechanisms.

Engineering Leadership and Projects

Team Lead, IUT Mars Rover Team (2022–2023)

- Designed and manufactured a fully functional autonomous rover for international rover competitions.
- Redesigned the suspension system from Rocker-Bogie to Four-Bar Linkage mechanism for improved terrain adaptability.
- Integrated in-wheel motor systems for smooth motion and 360° rotation capability.
- Designed and fabricated a 6-DoF robotic arm for precise manipulation.
- Developed an onboard science box for testing collected rock and soil samples.

Achievements

- ERC 2021 (On-site): Qualified with the 2nd highest score among 58 participating teams.
- ERC 2021 (Remote): Ranked 10th among 38 teams.
- IRC 2022: Qualified for the Final Round.
- IRDC 2022: Ranked 13th among 24 teams.

Chair, IMechE IUT Student Chapter (2022–2023)

- Organized technical sessions on engineering innovations and technical report writing.
- Arranged Speak Out for Engineering (SOFE) events and intra-IUT Robo Race competitions.

Vice President, IUT CAD Society (2022–2023)

- Organized CAD, modelling, and simulation workshops.
- Conducted technical sessions on SOLIDWORKS, ANSYS, and AutoCAD.
- Organized intra-IUT CAD competition Traction 2.0.

Awards and Achievements

- First Class with Honors Distinction in undergraduate studies, 2023.
- 1st Runner Up, Team CADmium, Thunder CAD (Inter-University CAD Competition), BUET Mecha Fest, 2023.
- Award of Achievement, Team Anirban, IUT Award Ceremony, 2023.
- Award of Achievement, Team CADmium, IUT Award Ceremony, 2023.
- Champion, Team CADmium, Techno CAD (Inter-University CAD Competition), AUST Mind Spark, 2022.
- Ranked 13th globally, International Rover Design Challenge (IRDC), 2022.
- Recipient of Government Scholarship in National Board Examination (HSC), 2018.
- Recipient of Government Scholarship in National Board Examination (SSC), 2016.

Volunteer Activities

- Organized Cezeri Lab Annual Project Showcase 2025.
- Organized ICCET-24, IUT.
- Organized 1st IUT International Conference on Mechanical, Materials and Production Engineering (ICMMPE) 2023.

References

Dr. Arafat Ahmed Bhuiyan
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Dr. Mohammad Monjurul Ehsan
Professor, Department of Mechanical and Production Engineering, IUT
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Chowdhury Sadid Alam
Assistant Professor, Department of Mechanical and Production Engineering, IUT
Email: chsadidalam@iut-dhaka.edu